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Custom Index

Custom Index and Manager Skill

We create a custom index to assess the skill of an active manager. We task code with selecting the strongest relationships between a manager and a set of ~100 commercial indexes that represent either liquid securities or known strategy betas. We do this across various lengths of lookback windows to compute index weights. **All weights are computed out of sample.** From these hypothetical portfolio weights we fashion a single time series that represents a non-deterministic benchmark. This should provide two key insights. First, the ability of a custom index to track manger performance should have an indirect relationship to manager fees. This is also the benefit of building the custom index out of sample. We attempt to build rock-stupid tracker, if the tracker can sufficiently duplicate a track record the fee for skill should be low. Second, the elements of a custom index should reflect the exposures expected by the investor, the manager's collateral, etc.

Custom Index Construction Telemetry

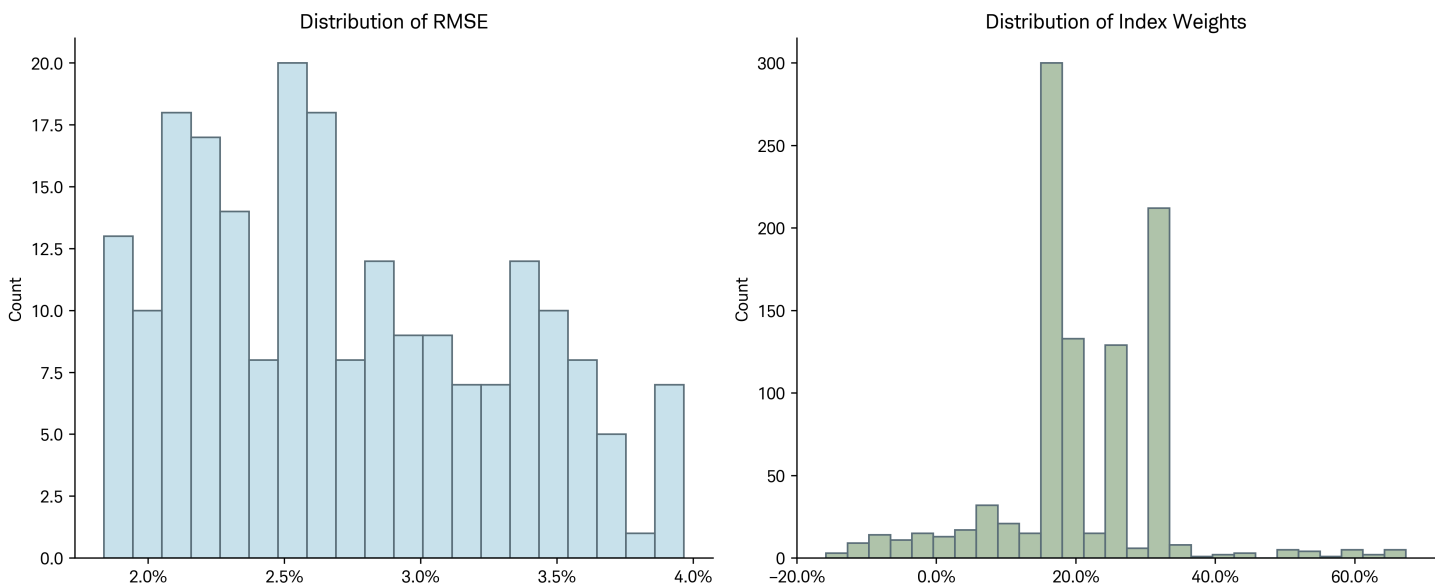


Figure 3.1 Some specifications for Fund A custom index. Left panel shows the distribution of RMSE computed for each lookback window. Right panel shows the distribution of index weights selected in each lookback window.

Count Best RMSE	
Window Length	
60	93
30	68
40	32
50	20

Table 3.1 Some specifications for Fund A custom index. For each rolling window, we attempt to fit to various window lengths. Table shows the count of optimal window lengths used to compute each set of weights.

Custom Index Construction Elements

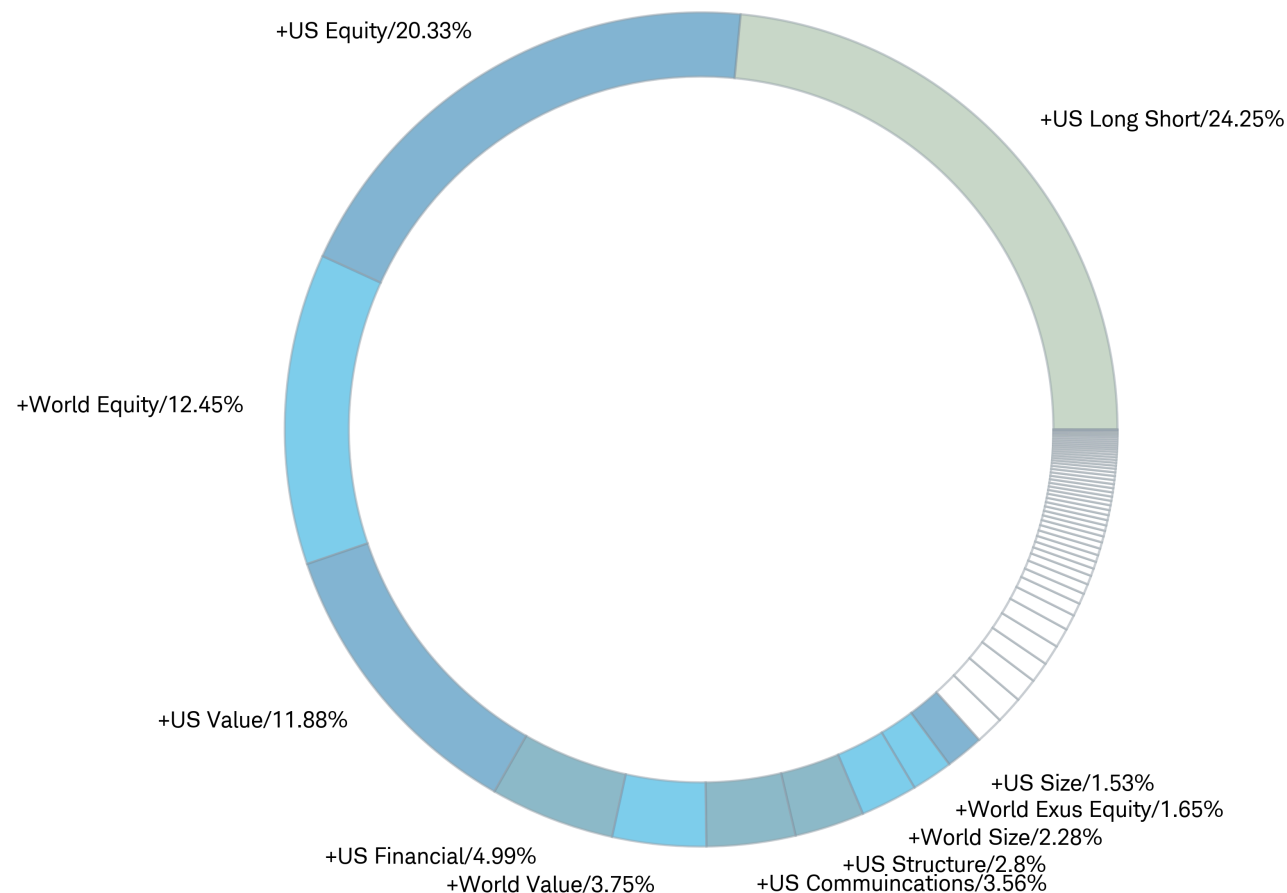


Figure 3.6 Summary of indexes used to create custom index for Fund A. Plus/minus prefix indicates long/short. Index weight is the percentage of all weights computed across all lookback windows. Weights less than 90% cumulative weight are excluded. Donut shows 89% of all weights computed.

	Index Weight	Cumulative Weight
(Long/Short) Index Name		
+US Long Short	24.3%	24.3%
+US Equity	20.3%	44.6%
+World Equity	12.4%	57.0%
+US Value	11.9%	68.9%
+US Financial	5.0%	73.9%
+World Value	3.8%	77.7%
+US Commuincations	3.6%	81.2%
+US Structure	2.8%	84.0%
+World Size	2.3%	86.3%
+World Exus Equity	1.6%	87.9%
+US Size	1.5%	89.5%

Table 3.2 Summary of indexes used to create custom index for Fund A. Plus/minus prefix indicates long/short. Index weight is the percentage of all weights computed across all lookback windows. All weights less than 3% are excluded. Table shows 81% of all weights computed.

Custom Index Performance Comparison

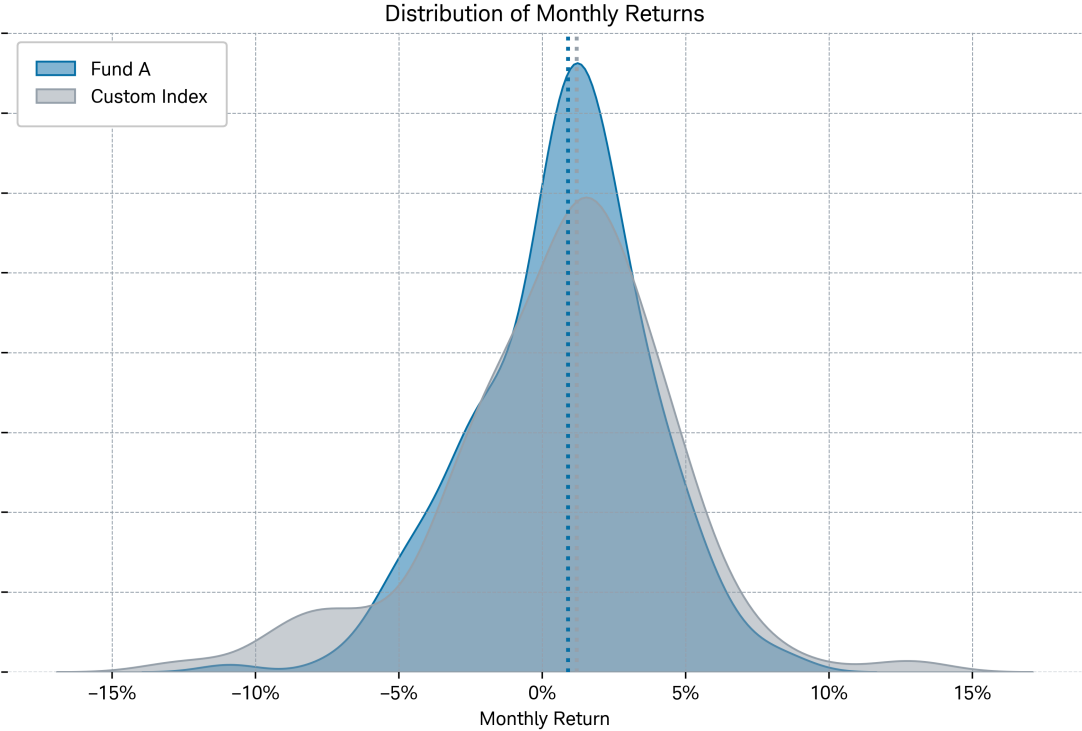


Figure 3.3 Distribution of monthly returns for Fund A and custom index. Dashed lines indicate distribution median. Intersection of return histories is 213 months from 07/31/2007 to 03/31/2025

Fund Name		Fund A	Custom Index
Lookback Length	Metric Name		
Full History	CAGR	8.27%	5.94%
	Ann.Volatility	10.34%	13.49%
	Arithmetic Mean	0.71%	0.56%
	Median	1.0%	1.0%
	Minimum	-11.0%	-13.0%
	Maximum	9.0%	13.0%
	Max Drawdown	23.4%	45.88%
	Max Drawup	441.27%	416.54%
	Average Upside	2.75%	3.1%
	Average Downside	-1.98%	-2.91%
	Ann.Volatility Upside	5.78%	7.43%
	Ann.Volatility Downside	7.13%	10.25%
	Var (95)	4.4%	7.0%
	cVar (5)	5.91%	8.85%
	Skew	-0.39	-0.51
	Kurtosis	0.67	1.5
	Batting Average	0.57	0.58
	Calmar Ratio	0.35	0.13
	Sharpe Ratio (rf=True)	0.64	0.32
	Sortino Ratio (rf=True)	0.93	0.42

Table 3.3 Performance and risk statistics for Fund A and custom index. Intersection of return histories is 213 months from 07/31/2007 to 03/31/2025

Custom Index as Benchmark

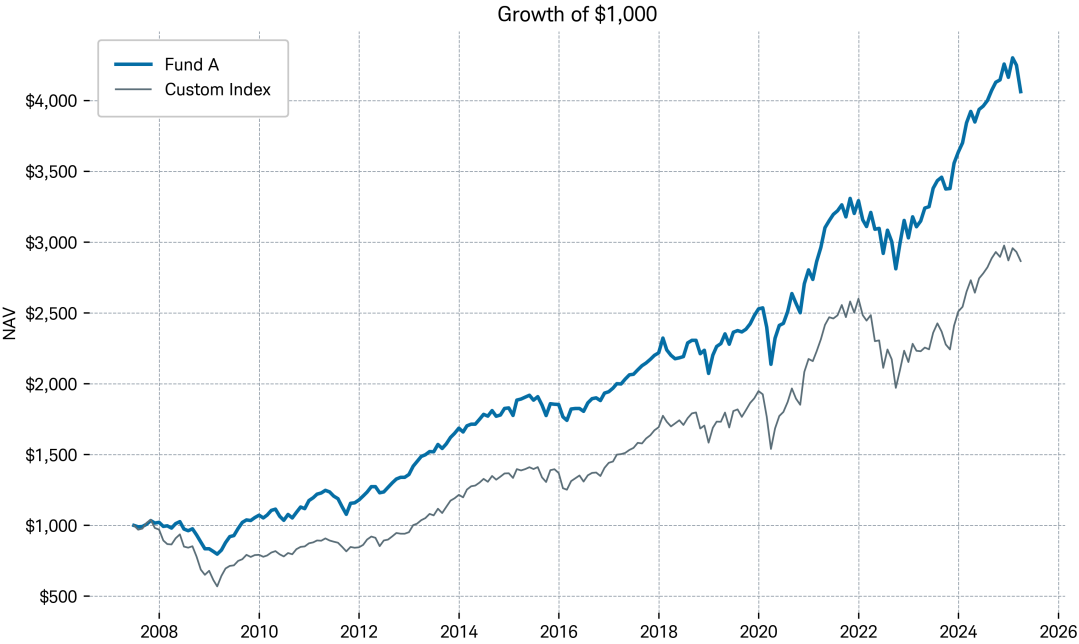


Figure 3.4 Growth of \$1,000 for Fund A and custom index. Dashed lines indicate distribution median. Intersection of return histories is 213 months from 07/31/2007 to 03/31/2025

vs. Benchmark	
Alpha (%)	0.31
Beta	0.68
Active Return (%)	2.11
Tracking Error	6.30
Information Ratio	0.33
Batting Average	0.47
Upside Capture	121.86
Downside Capture	117.58

Table 3.4 Performance and risk statistics for Fund A and custom index. Intersection of return histories is 213 months from 07/31/2007 to 03/31/2025

Custom Index as Benchmark

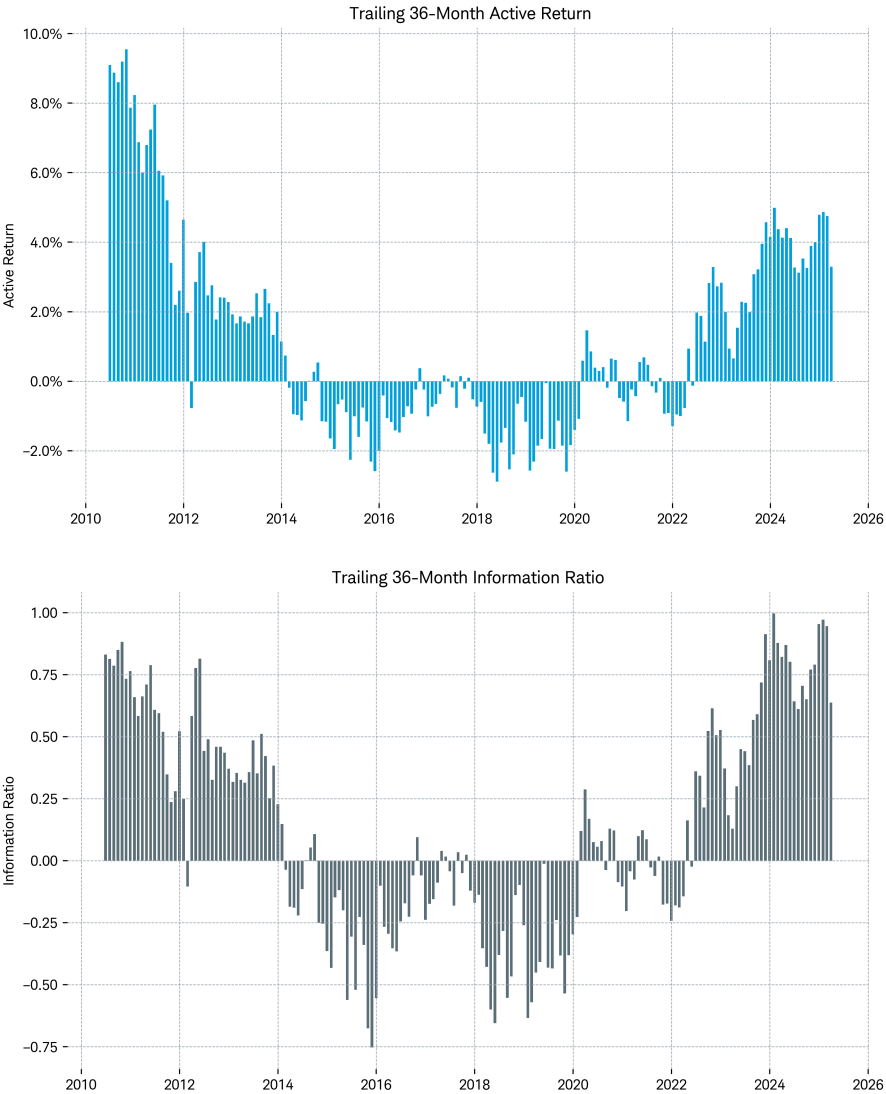
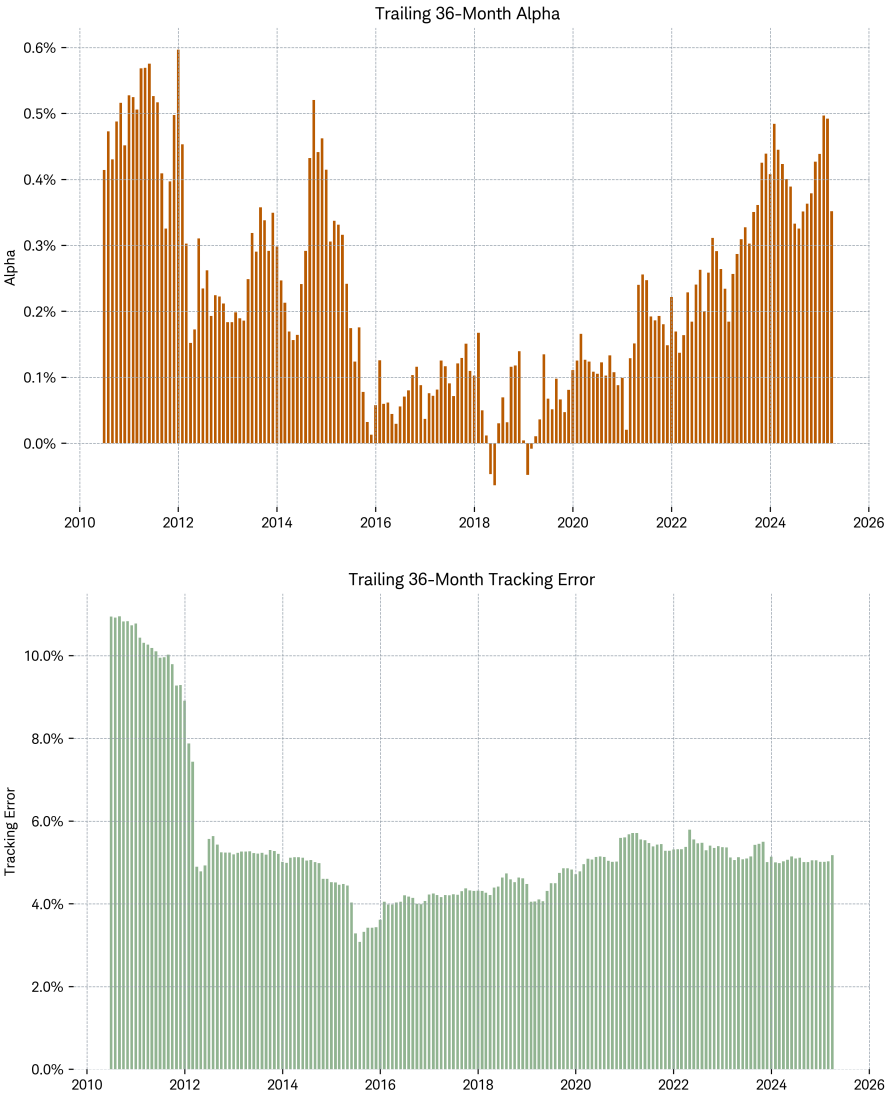


Figure 3.5-8 Various trailing benchmark metrics for Fund A versus custom index. Intersection of return histories is 213 months from 07/31/2007 to 03/31/2025

End